

PAVAN KUMAR KAMARAJUGADDA

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PROFESSIONAL SUMMARY

Senior Software Engineer with ~5 years of building production backend systems, distributed APIs, and cloud-native platforms. Strong track record shipping at scale — from microservices and API gateway design to async messaging and infrastructure automation. Over the past year, extended that foundation into the AI/ML space, delivering GenAI features inside a healthcare enterprise using **LangChain**, **FastAPI**, **GCP**, and LLM APIs. Comfortable owning systems end-to-end — from architecture and code to deployment. Looking for a senior SDE role where engineering craft matters.

TECHNICAL SKILLS

Languages	Python, Node.js, JavaScript / TypeScript
Frameworks	FastAPI, Express.js, React.js, Redux, GraphQL
Architecture	Microservices, REST, gRPC, API Gateway design, Event-driven (RabbitMQ)
Cloud & Infra	AWS (S3, Lambda), GCP (Cloud Run, Vertex AI, BigQuery, GCS), Docker, Kubernetes, GitHub Actions
Databases	MySQL, MongoDB, Redis, BigQuery
AI / GenAI	LangChain, LlamaIndex, OpenAI API, Anthropic API, RAG Pipelines, Pinecone, FAISS
Testing & Docs	PyTest, Postman, Swagger / OpenAPI, MLflow

PROFESSIONAL EXPERIENCE

Senior Software Engineer | HCA Healthcare India | Hyderabad | May 2025 – Present

- Designed and shipped an end-to-end speech-to-report pipeline for emergency physicians: integrated **MedASR** for clinical speech transcription with **Med-Gemma** for contextual understanding, orchestrated via **LangChain**. Built the **FastAPI** backend, containerized with **Docker**, and deployed on **Google Cloud Run** with artifacts in **GCS**. Cut physician documentation time by ~1 hour per shift.
- Built a Named Entity Recognition service on top of **Google Vertex AI** (serving **Med-Gemma**) that auto-detects and highlights critical medical terms inline within provider reports. Exposed via **REST API** and integrated into the document review workflow, cutting average reviewer clearance time significantly.
- Part of team Converge — built a shared enterprise API layer using **FastAPI**, documented with **Swagger/OpenAPI**, deployed on **Kubernetes**, backed by **BigQuery** and upstream services. Eliminated duplicated data access logic across downstream services, reducing new integration effort and enforcing consistent data access patterns org wide.

Senior Developer | DevRabbit IT Solutions Pvt Ltd | Hyderabad | Dec 2022 – May 2025

SuiteAmerica — Corporate Housing Platform

- Suite America's property listing service was entirely US-hosted, and European clients were experiencing noticeable lag when browsing and booking properties. Extended the infrastructure to a dual-region setup on **AWS**, deploying the application in both US and EU regions and using **Route 53 for latency-based routing** to direct each user to the nearest region. Writes remained centralized in the US database with cross-region replication pushing data to the EU, keeping the system consistent while reads stayed fast.
- Property listing pages are read-heavy by nature — every client browsing available units hits the same data repeatedly. Layered in **Redis** caching for property details and availability data, reducing database load and improving page response times for users across both regions.
- Owned full-stack feature development across the platform — new booking flows, property listing screens, and the corresponding backend changes — using **React.js** on the frontend and **Node.js / REST APIs** on the backend.

Restaurant Task Management Platform

- Built a task management system for restaurant teams from the ground up — owners can create tasks, assign them to staff, and tag people so tasks surface in the right feeds. Supported sub-tasks as checklists

within each task, and proof-of-work attachments (photos, documents, signatures) so completion was verifiable, not just self-reported.

- Designed and built a standalone **notifications microservice** to handle multi-channel delivery (email, SMS, in-app via **Firebase**) — decoupled it intentionally so other services could reuse it without duplicating delivery logic. Backed it with **RabbitMQ** and **AWS SQS** for async fanout, with the system reliably handling 10,000+ concurrent users.
- Deployed the entire platform on a **Kubernetes** cluster, enabling independent scaling per service based on actual load — the notification service and task API could scale at their own rates without touching each other.

Software Developer | Code Connex Pvt. Ltd. | Hyderabad | Apr 2022 – Dec 2022

- Several REST endpoints were over-fetching, bloating payloads and dragging down frontend load times. Introduced **GraphQL** as a middleware aggregation layer on top of **Node.js**, replacing multiple chatty REST calls with precise field-level queries. API payload size dropped by 50% and data retrieval speed improved noticeably.
- Developed and optimized high-performance web applications using **React.js**, **Redux**, and **Node.js**, delivering measurable improvements in UI responsiveness and end-user experience.

Project Engineer | Wipro Ltd. | Bengaluru | Jun 2018 – Mar 2019

- The finance team was manually keying invoice data into internal systems — slow and error-prone at volume. Built an automated extraction system using **Python**, **OpenCV**, and **Tesseract OCR** to detect, extract, and normalize invoice fields, then wrapped it in a full-stack **React.js / Express.js / MongoDB** document management app for end-to-end ingestion and retrieval. Cut manual data entry workload by 60% at ~90% accuracy.

EDUCATION

M.Sc. in Exploration Geophysics — Indian Institute of Technology (IIT) Kharagpur | 2013 – 2018 |

CGPA: 7.96 / 10